Research preprint

An affine Hankel construction for $\zeta(7)$

Lean-verified algebra and conditional irrationality

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Abstract

We describe a rational Hankel determinant family using the kernel $D_{3n}(t)^6/D_{40n}(t)$ and an order-seven affine functional. For this explicitly defined family $F_n \in \mathbb{Q}[X]$, the accompanying Lean package proves natdeg $F_n \le 37n$, positivity of its factorial scalar, and existence of fixed positive multipliers giving integer polynomials. It also proves a conditional irrationality theorem for the real part of Mathlib's actual Riemann zeta function at 7. If the same family satisfies a positive analytic bound with quadratic rate U, admits fixed integer normalizations with quadratic growth rate A, and $A + U < 0$, then its normalized values have quadratic exponential decay and $\zeta(7)$ is irrational. We give the algebraic proofs, the exact quantified hypotheses, and the integer-denominator contradiction. The supporting compilation is reported in a supplied native Lean 4.33.0 audit transcript. The analytic estimate, normalization-growth estimate, and a compatible negative gap remain unproved; no unconditional irrationality result is asserted.

Keywords. Riemann zeta function; irrationality criterion; Hankel determinant; denominator clearing; Lean formalization.

1 Main result and scope

Let

$$x_7 := \mathrm{Re}\big(\zeta(7)\big), \qquad K_n := 40n. \tag{1.1}$$

Here the formal definition is `(riemannZeta (7 : ℂ)).re`; in the usual real notation $x_7 = \zeta(7)$. The target is a particular real number, not an abstract parameter assumed to equal a zeta value. The family $F = (F_n)$ is defined in Section 2 by finite rational polynomial operations.

The result below is the reduced endpoint of `ZETA7_DERIVED_ALGEBRA_v0_1_2_FASTCORE` [1]. Write $\mathsf{AN}(F, x, U)$ and $\mathsf{AR}(F, A)$ for the following properties; all equalities between polynomials over different coefficient rings are interpreted through the canonical embeddings.

$$\begin{aligned} \mathsf{AN}(F,x,U): \quad & \forall \varepsilon > 0\ \exists N_\varepsilon\ \forall n \ge N_\varepsilon : \\ & F_n(x) > 0, \qquad \log F_n(x) \le (U+\varepsilon)K_n^2; \end{aligned} \tag{1.2}$$

$$\begin{aligned} \mathsf{AR}(F,A): \quad & \exists m : \mathbb{N} \to \mathbb{Q}\ \exists Q : \mathbb{N} \to \mathbb{Z}[X]\ \forall \varepsilon > 0 \\ & \exists M_\varepsilon\ \forall n \ge M_\varepsilon : \quad m_n > 0, \quad Q_n = m_n F_n, \\ & \qquad\qquad \log m_n \le (A+\varepsilon)K_n^2. \end{aligned} \tag{1.3}$$

The witnesses m and Q in (1.3) are chosen *before* ε. Thus one normalization family must satisfy every tolerance. The real logarithm in (1.3) is applied after embedding m_n into $\mathbb{R}$.

Theorem 1.1 (Conditional irrationality for the fixed candidate)**.** *For the family F of Section 2, let $A, U \in \mathbb{R}$. Assume*

$$\mathsf{AN}(F, x_7, U), \qquad \mathsf{AR}(F, A), \qquad A + U < 0. \tag{1.4}$$

Then x_7 is irrational. More precisely, with

$$\delta := -(A+U) > 0, \qquad c := 800\delta, \tag{1.5}$$

there are integer polynomials Q_n which, for all sufficiently large n, satisfy

$$\mathrm{natdeg}\, Q_n \le 37n, \qquad 0 < Q_n(x_7) \le \exp(-cn^2), \qquad Q_n = m_n F_n \quad (m_n \in \mathbb{Q}_{>0}). \tag{1.6}$$

The proof is given in Sections 5–6; Appendix A identifies the corresponding Lean declarations. No additional divergence hypothesis is required. The degree bound needed by the proof is established for the candidate, rather than assumed. Existence of some integer normalization is also established, but does not supply the growth assertion in (1.3).

The role of this note is to present the exact conditional result supported by the source package and supplied audit [1, 2]. The generic integer-polynomial contradiction is not presented as a new irrationality criterion. The specific content here is the candidate definition, the proved algebraic components, and their binding to a precisely stated analytic–arithmetic interface.

2 The explicitly defined rational family

We distinguish the rational-function variable t from the output indeterminate X. Put

$$K = 40n, \qquad N = 3n, \qquad h = 37n, \qquad D_m(t) = \prod_{j=1}^{m}(t + j^2). \tag{2.1}$$

Products over empty index sets have value 1.

2.1 Order-seven moments and pole values

Let B_k be the Bernoulli numbers with $B_1 = -1/2$. For $e \geq 0$, set

$$a_7(e) = \frac{(-1)^e B_{2e+2}}{720} \prod_{r=3}^{7}(2e + r). \tag{2.2}$$

For a rational polynomial $P(t) = \sum_e p_e t^e$, define

$$\mathcal{L}_7(P) := \sum_e p_e a_7(e) \in \mathbb{Q}. \tag{2.3}$$

For $j \geq 1$, put

$$H_j^{(7)} = \sum_{v=1}^{j} v^{-7}, \qquad V_{7,j}(X) = j^6\big(X - H_j^{(7)}\big) + \frac{1}{2j} - \frac{1}{6}. \tag{2.4}$$

All coefficients in these formulas are rational. In particular, $V_{7,j}$ is affine in X.

2.2 Finite definition of the functional

For $P \in \mathbb{Q}[t]$, let $\mathrm{quo}(P, D_K)$ denote its polynomial quotient on division by the monic polynomial D_K. Define

$$\begin{aligned}\mathcal{M}_K(P; X) := {} & \mathcal{L}_7(\mathrm{quo}(P, D_K)) \\ & + \sum_{j=1}^{K} \frac{P(-j^2)}{D_K'(-j^2)}\, V_{7,j}(X).\end{aligned} \tag{2.5}$$

The roots $-1^2, \ldots, -K^2$ are distinct; for $1 \leq j \leq K$,

$$D_K'(-j^2) = \prod_{\substack{1 \leq v \leq K \\ v \neq j}} (v^2 - j^2) \neq 0. \tag{2.6}$$

Thus (2.5) has the usual polynomial-division and simple-residue interpretation for P/D_K. Crucially, the finite formula (2.5) itself is the definition in the source. No analytic integral identity is imported to define it.

2.3 The matrix and its scalar normalization

Define the polynomial matrix and determinant by

$$G_n(X)_{ij} := \mathcal{M}_{40n}\left(D_{3n}(t)^6 t^{i+j}; X\right), \quad 0 \le i, j < h, \qquad \Delta_n(X) := \det G_n(X). \tag{2.7}$$

The entry depends on $i + j$, so the matrix is Hankel. Its rational-function kernel is

$$\frac{D_{3n}(t)^6}{D_{40n}(t)}. \tag{2.8}$$

For $n \ge 1$, set

$$S_n = \frac{(K!)^{2h}\, 4^{h-1}}{(N!)^{12h} \prod_{i=1}^{h-1} ((2i)!)^2}, \qquad F_n(X) := S_n \Delta_n(X) \in \mathbb{Q}[X]. \tag{2.9}$$

The Lean definitions also cover $n = 0$: natural subtraction truncates at zero, the matrix is empty, and $S_0 = \Delta_0 = F_0 = 1$. This convention does not affect any eventual hypothesis or conclusion.

The construction retains the sixth-power kernel and the factorial exponent $12h$. It does not use the eighth-power kernel of the previously uploaded claimed $\zeta(7)$ proof. Neither that paper's main theorem nor any external $\zeta(5)$ theorem is imported by these modules [1].

3 The degree bound and factorial positivity

For a polynomial P, use the natural degree natdeg P and the convention natdeg $0 = 0$, matching Lean's `Polynomial.natDegree`. This convention allows an upper bound without first proving that P is nonzero.

Lemma 3.1 (Affine entries)**.** *For every K and every $P \in \mathbb{Q}[t]$,*

$$\operatorname{natdeg}_X \mathcal{M}_K(P; X) \le 1. \tag{3.1}$$

In particular, every entry of G_n has natural degree at most one.

Proof. The first term in (2.5) is constant in X. Every term of the finite sum is a rational constant times the affine polynomial $V_{7,j}(X)$. A finite sum cannot increase the maximum of the natural degrees. Equation (2.7) gives the conclusion for G_n. □

Lemma 3.2 (Determinant of an affine matrix)**.** *If M is an $h \times h$ matrix over $\mathbb{Q}[X]$ and every entry has natural degree at most one, then* natdeg det $M \le h$.

Proof. By the Leibniz formula,

$$\det M = \sum_{\sigma \in \mathfrak{S}_h} \operatorname{sgn}(\sigma) \prod_{i=0}^{h-1} M_{\sigma(i),i}. \tag{3.2}$$

For each permutation, the product has natural degree at most $\sum_i \operatorname{natdeg} M_{\sigma(i),i} \le h$. A sign does not change natural degree, and the degree of the sum is bounded by the largest summand degree. For $h = 0$, the determinant is 1 and the same bound holds. □

Proposition 3.3 (Degree input discharged)**.** *For every $n \in \mathbb{N}$,*

$$\operatorname{natdeg} \Delta_n \le 37n, \qquad \operatorname{natdeg} F_n \le 37n. \tag{3.3}$$

Proof. Apply Lemmas 3.1 and 3.2 with $h = 37n$. Multiplication by the constant polynomial S_n cannot increase natural degree. □

Proposition 3.4 (Positivity of the explicit scalar)**.** *For every $n \in \mathbb{N}$, $S_n > 0$.*

Proof. Every factorial in (2.9) is positive, as is 4. Powers and finite products of these quantities are positive. The denominator is nonzero, so their quotient is positive; the convention $S_0 = 1$ treats $n = 0$. □

Proposition 3.3 is an upper bound, not an exact degree formula. Neither it nor Proposition 3.4 establishes $\Delta_n(x_7) > 0$. The latter is still part of the analytic obligation through $F_n(x_7) > 0$.

4 Integer normalization without a growth estimate

Lemma 4.1 (Positive denominator clearing)**.** *For every $P \in \mathbb{Q}[X]$, there exist $d \in \mathbb{N}_{>0}$ and $Q \in \mathbb{Z}[X]$ such that*

$$\iota(Q) = dP \quad \text{in } \mathbb{Q}[X], \tag{4.1}$$

where $\iota : \mathbb{Z}[X] \hookrightarrow \mathbb{Q}[X]$ is coefficientwise inclusion.

Proof. For a monomial cX^k, write $c = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{N}_{>0}$. Then $b(cX^k) = aX^k$ has integer coefficients. For the inductive addition step, suppose $\iota(P_0) = ap$ and $\iota(Q_0) = bq$ with positive integers a, b. Then

$$\iota(bP_0 + aQ_0) = ab(p + q). \tag{4.2}$$

Induction on a polynomial as a finite sum of monomials proves the claim. □

Proposition 4.2 (Fixed normalization witnesses exist)**.** *There exist sequences $m : \mathbb{N} \to \mathbb{Q}$ and $Q : \mathbb{N} \to \mathbb{Z}[X]$ such that, for every n,*

$$m_n > 0, \qquad \iota(Q_n) = m_n F_n, \qquad \operatorname{natdeg} Q_n \le 37n. \tag{4.3}$$

The multipliers in this existence proof may be chosen to be positive integers.

Proof. Apply Lemma 4.1 to each F_n and choose a corresponding pair d_n, Q_n. Set $m_n = d_n$ as a rational number. The coefficient embedding is injective, and multiplication by a nonzero scalar preserves natural degree. Hence

$$\operatorname{natdeg} Q_n = \operatorname{natdeg} \iota(Q_n) = \operatorname{natdeg}(m_n F_n) = \operatorname{natdeg} F_n \le 37n. \tag{4.4}$$

The fixed sequences in the source are supplied using classical choice. The same degree argument applies to any positive rational multiplier satisfying the polynomial identity, not only to this choice. □

Remark 4.3 (Existence is not asymptotic control). Proposition 4.2 does not prove (1.3) for any specified finite A. It imposes no upper bound on $\log m_n / K_n^2$. Moreover, (1.3) may use a different fixed pair of sequences from the one chosen in the existence proof. Positivity, integrality, and the growth estimate in (1.3) must concern that same pair. No primitivity or optimality of a normalizer is asserted or needed.

5 The integer-polynomial irrationality criterion

This section isolates the elementary contradiction used by the formal endpoint. It does not require a determinant representation.

Lemma 5.1 (Denominator lower bound)**.** *Let $P \in \mathbb{Z}[X]$, let $a \in \mathbb{Z}$, $b \in \mathbb{N}_{>0}$, and suppose* natdeg $P \le k$ *and $P(a/b) > 0$. Then*

$$b^k P(a/b) \in \mathbb{Z}_{>0}, \qquad P(a/b) \ge b^{-k}, \qquad -\log P(a/b) \le k \log b. \tag{5.1}$$

Proof. Writing $P(X) = \sum_{i=0}^{k} a_i X^i$ with $a_i \in \mathbb{Z}$, one has

$$b^k P(a/b) = \sum_{i=0}^{k} a_i a^i b^{k-i} \in \mathbb{Z}. \tag{5.2}$$

The left side is positive and therefore at least 1. Division by $b^k > 0$ gives the second claim, and taking logarithms gives the third. □

Definition 5.2 (Logarithmic divergence)**.** For a real x and a sequence $Q_n \in \mathbb{Z}[X]$, the positive-index form of logarithmic divergence is

$$\forall C \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \ge \max\{N, 1\} : \quad C < \frac{-\log Q_n(x)}{n}. \tag{5.3}$$

Whenever used for irrationality, positivity of $Q_n(x)$ is supplied separately for sufficiently large n. The source's `LogDivergence` uses $n \ge N$; taking $N \ge 1$ gives the displayed equivalent eventual form.

Theorem 5.3 (Generic conditional criterion)**.** *Let $D \in \mathbb{N}$ and $Q_n \in \mathbb{Z}[X]$. Suppose, for all sufficiently large n,*

$$\text{natdeg}\, Q_n \le Dn, \qquad Q_n(x) > 0, \tag{5.4}$$

and suppose (5.3) *holds. Then x is irrational.*

Proof. Assume $x = a/b$ with $a \in \mathbb{Z}$ and $b \in \mathbb{N}_{>0}$. Choose a positive n beyond the structural threshold for which $-\log Q_n(x)/n > D \log b$. Lemma 5.1 with $k = Dn$ gives the opposite inequality. This is a contradiction. □

Corollary 5.4 (Quadratic decay suffices)**.** *If $c > 0$ and, eventually,*

$$\text{natdeg}\, Q_n \le Dn, \qquad 0 < Q_n(x) \le e^{-cn^2}, \tag{5.5}$$

then x is irrational.

Proof. The estimate gives $-\log Q_n(x)/n \ge cn$ for sufficiently large positive n. The right side becomes arbitrarily large, so Theorem 5.3 applies. □

6 Combining the two growth inputs

We now prove Theorem 1.1, keeping the witnesses and error parameters explicit.

Proof of Theorem 1.1. Let $\delta = -(A + U) > 0$ and choose $\varepsilon = \delta/4$. Apply (1.2) at this tolerance. From (1.3), first choose the fixed sequences m, Q, then apply their growth assertion at the same tolerance. Beyond the maximum of the two thresholds and 1, we have

$$F_n(x_7) > 0, \qquad \log F_n(x_7) \le (U + \delta/4)K_n^2, \tag{6.1}$$

$$m_n > 0, \qquad \log m_n \le (A + \delta/4)K_n^2, \tag{6.2}$$

$$\iota(Q_n) = m_n F_n. \tag{6.3}$$

The polynomial identity (6.3), after evaluation in $\mathbb{R}$, gives

$$Q_n(x_7) = m_n F_n(x_7) > 0. \tag{6.4}$$

This equality uses coefficient maps and evaluation; it is not an additional numerical hypothesis. Because both factors are positive,

$$\begin{aligned} \log Q_n(x_7) &= \log m_n + \log F_n(x_7) \\ &\le (A + U + \delta/2)K_n^2 \\ &= -\frac{\delta}{2}K_n^2 = -800\delta n^2. \end{aligned} \tag{6.5}$$

Proposition 3.3, injectivity of ι, and $m_n \neq 0$ give $\operatorname{natdeg} Q_n \le 37n$. Exponentiating (6.5) gives (1.6) with $c = 800\delta > 0$. Apply Corollary 5.4 with $D = 37$. □

6.1 Direct contradiction at a hypothetical rational value

Equivalently, if $x_7 = a/b$ with $b > 0$, then for every sufficiently large n,

$$1 \le b^{37n} Q_n(a/b) \le \exp\left(37n \log b - 800\delta n^2\right). \tag{6.6}$$

The exponent is negative whenever $n > 37 \log b/(800\delta)$ and $n > 0$. Taking n also beyond the estimate threshold makes the right side strictly less than 1, a contradiction.

6.2 What is derived, and what is an input

The two growth blocks and the strict inequality $A + U < 0$ derive both quadratic exponential decay and logarithmic divergence:

$$\boxed{\mathsf{AN}(F, x_7, U) \wedge \mathsf{AR}(F, A) \wedge (A + U < 0)} \implies (1.6) \implies x_7 \notin \mathbb{Q}. \tag{6.7}$$

The constants A and U are free parameters of the theorem. The package does not supply values for which both input blocks hold and the strict gap is negative. In particular, the last inequality in (1.4) is not obtained by merely choosing numbers: the corresponding analytic and arithmetic statements must also be proved for those numbers.

7 Remaining mathematical obligations

After the proved algebra is substituted, the endpoint has two estimate blocks and one compatibility condition.

Obligation	**Statement still required for the fixed candidate**
Analytic	Eventual strict positivity of $F_n(x_7)$ and the upper bound (1.2) for a real constant U.
Arithmetic	A single pair m, Q satisfying the polynomial normalization identity and the growth estimate (1.3) for a real constant A.
Strict gap	The same constants satisfy $A + U < 0$.

The scalar theorem $S_n > 0$ is insufficient for the first obligation, since $F_n = S_n\Delta_n$ and the sign of the determinant value is not settled by scalar positivity. The existence theorem in Section 4 is insufficient for the second obligation, since no asymptotic size estimate follows from clearing finitely many denominators at each index.

7.1 A regression check on earlier proposed constants

The architecture module includes the exact rational identity

$$\frac{44617}{19200} - \frac{2733991}{2000000} = \frac{11481679}{12000000} > 0. \tag{7.1}$$

Its declarations prove arithmetic about these constants only. They do not prove that the constants bound this candidate. Even if the associated proposed estimates were established, this pair would not satisfy the strict gap required by Theorem 1.1. A positive combined upper bound is also not a lower bound: it does not prove that the actual normalized values grow.

These two declarations belong to the compiled architecture module and were included in the earlier architecture audit supplied in the conversation. The later v0.1.2 audit primarily prints the newly derived algebraic declarations and the reduced endpoint [2].

7.2 Relation to an order-five-style proof architecture

The intended organization is a linear degree bound, a positive analytic estimate, an integer normalization with a growth bound, and a negative sum of rates. The current package formalizes that organization for its order-seven candidate and discharges the degree input. This structural analogy is not a transfer theorem for analytic or arithmetic estimates. No conclusion about the correctness or completeness of a separate $\zeta(5)$ proof is needed anywhere in this note.

8 Formalization and supplied verification record

The source archive contains six mathematical modules, with `Audit.lean` importing the reduced endpoint. The dependency chain is

$$\begin{gathered}\{\text{Criterion}, \text{Candidate}\} \longrightarrow \text{Architecture},\\ \text{Architecture} \longrightarrow \text{Degree} \longrightarrow \text{Normalization} \longrightarrow \text{ReducedClosure}.\end{gathered} \tag{8.1}$$

Here each short label denotes the corresponding `ZetaSeven`*Label*`.lean` module.

The supplied execution transcript reports successful compilation of all six modules, successful execution of `Audit.lean`, and `PROJECT_BUILD=PASS` and `RUN=PASS` [2]. It identifies the compiler as

```
Lean 4.33.0
x86_64-w64-windows-gnu
commit d8b18978322de05a8f3dba51ef03cf5461676c17
```

The runner uses the pre-existing native toolchain and cached Mathlib, with 16 jobs in the supplied run. It does not invoke Lake or download dependencies during that run. This is a cache-reusing execution setup, not a self-contained distribution of the complete dependency environment.

For the declarations printed in the successful v0.1.2 audit, the reported dependencies are

```
[propext, Classical.choice, Quot.sound]
```

and no `sorryAx` is reported. The source also contains a static proof-escape scan. That scan is an auxiliary check; the theorem types, proof compilation, and printed dependencies are the relevant evidence.

Remark 8.1 (Provenance of the verification claim). The compilation status stated here is supported by the user-supplied native Windows transcript, together with inspection of the exact source archive. Preparing this preprint did not independently rebuild Lean, Mathlib, or the project. The archive's README records its earlier pre-run status; the supplied successful execution occurred subsequently. The local theorem being reported is the conditional endpoint below, not an assumption-free irrationality declaration.

Statement	**Recorded status**
Specific F_n and actual zeta-value binding	Defined in the source
natdeg $F_n \le 37n$	Proved; supplied run passed
$S_n > 0$	Proved; supplied run passed
Fixed integer-normalization witnesses exist	Proved; supplied run passed
Normalized degree bound	Proved; supplied run passed
Two inputs plus gap imply irrationality	Proved conditionally
Determinant positivity and analytic growth	Retained hypothesis
Normalizer growth and a compatible strict gap	Retained hypotheses

9 Conclusion

For the explicitly defined order-seven Hankel family, the supplied formalization establishes the algebraic degree bound, positivity of the factorial scalar, and fixed integer-normalization witnesses. These results are sufficient to remove the degree hypothesis from the original three-block conditional architecture, but not to establish the two growth blocks.

The resulting theorem has the exact form

$$\boxed{\mathsf{AN}(F, x_7, U) \ \wedge \ \mathsf{AR}(F, A) \ \wedge \ (A + U < 0) \implies \mathrm{Irrational}(x_7).} \tag{9.1}$$

The remaining mathematical task is to establish both estimates for this same candidate with a compatible negative gap. No claim that those estimates have already been proved is part of the present result.

A Lean interfaces and theorem correspondence

The following excerpts are from the v0.1.2 source. Names are retained to make the mathematical statements directly locatable [1].

A.1 Growth inputs and fixed witnesses

Equations (1.2) and (1.3) reproduce the quantifier order of `AnalyticInput` and `ArithmeticInput`. The latter begins $\exists m\, \exists Q\, \forall \varepsilon > 0$, not $\forall \varepsilon > 0\, \exists m\, \exists Q$. The source sets $K_n = 40n$ through `Kr`.

A.2 Reduced endpoint

```
theorem ZETA7_DEGREE_DISCHARGED
    (A U : ℝ)
    (ANALYTIC_PREMISE :
      AnalyticInput ZetaSevenCandidate.F zetaSeven U)
    (ARITHMETIC_PREMISE :
      ArithmeticInput ZetaSevenCandidate.F A)
    (GAP_PREMISE : A + U < 0) :
    Irrational zetaSeven := by
  exact irrational_of_three_inputs
    ZetaSevenCandidate.F zetaSeven A U
    ZetaSevenDegree.degree_input
    ANALYTIC_PREMISE ARITHMETIC_PREMISE GAP_PREMISE
```

The mathematical hypothesis `DEGREE_PREMISE` is absent. The supplied proof `ZetaSevenDegree.degree_input` replaces it. The analytic, arithmetic, and gap hypotheses remain visible.

A.3 Principal declaration map

In the following table, each name is prefixed by the indicated namespace.

Mathematical result	Lean declaration
Affine functional	`ZetaSevenDegree.muX_natDegree_le`
Affine determinant degree	`ZetaSevenDegree.affine_det_natDegree_le`
Candidate degree bound	`ZetaSevenDegree.degree_F`
Scalar positivity	`ZetaSevenDegree.S_positive`
Denominator clearing	`ZetaSevenNormalization.positive_integer_multiple`
Fixed witnesses	`ZetaSevenNormalization.integer_normalization_exists`
Normalized degree bound	`ZetaSevenNormalization.normalized_degree_le`
Integer criterion	`ZetaSevenCriterion.irrational_of_log_divergence`
Combined logarithmic bound	`ZetaSevenArchitecture.three_inputs_quadratic_estimate`
Exponential bound	`ZetaSevenArchitecture.quadratic_estimate_implies_exp_bound`
Reduced endpoint	`ZetaSevenReducedClosure.ZETA7_DEGREE_DISCHARGED`

B Source identity and reproduction

The source archive inspected for this note is

`ZETA7_DERIVED_ALGEBRA_v0_1_2_FASTCORE.zip`.

Its SHA-256 digest, computed from the available archive, is

```
b945c9c530a12c271b696c57214e52a198
56545222d3446aa81546c8b0727369
```

The two lines are one continuous 64-character digest. The archive also contains a per-file SHA-256 manifest, proof sources, the runner, an audit protocol, and derivation notes. These identify the inspected source; they do not replace a build against the reader's dependency cache.

On the established Windows installation, the command sequence reported in the supplied successful run is

```
cd /d "%USERPROFILE%\Downloads"
tar -xf "ZETA7_DERIVED_ALGEBRA_v0_1_2_FASTCORE.zip"
call "ZETA7_DERIVED_ALGEBRA_v0_1_2_FASTCORE\run.cmd"
```

The default cache root is

```
%USERPROFILE%\lean-native\awpi_phi_rh_lean_sieved_v0_1_1
```

The archive is not a standalone Mathlib distribution. Reproduction requires the matching native toolchain and already available compatible cache. The runner permits the documented environment-variable overrides contained in `run.cmd`.

The relevant final status lines in the supplied transcript include

```
DEGREE_BOUND=PROVED
DEGREE_PREMISE=DISCHARGED
INTEGER_NORMALIZATION_EXISTS=PROVED
REMAINING_ESTIMATE_BLOCKS=2
DETERMINANT_POSITIVITY=STILL_IN_ANALYTIC_PREMISE
NORMALIZER_GROWTH=NOT_PROVED
STRICT_GAP_PREMISE=EXPLICIT_UNPROVED
PAPER_THEOREM_1_2=NOT_USED
UNCONDITIONAL_IRRATIONALITY=NOT_PROVED
PROJECT_BUILD=PASS
RUN=PASS
```

These status lines summarize the signatures; the preceding printed theorem declarations establish which assumptions remain explicit.

References

[1] *ZETA7 Derived Algebra*, version 0.1.2 FASTCORE. Project source archive, 20 October 2026. Six mathematical modules, `Audit.lean`, the native runner, derivation notes, and a per-file SHA-256 manifest. Exact declarations are mapped in Appendix A; archive identity is recorded in Appendix B.

[2] *Native Windows Lean 4.33.0 build and axiom-audit transcripts*. User-supplied project records, 20 October 2026: the successful v0.1.2 run and earlier architecture audit. These records print theorem signatures, axiom dependencies, and remaining hypotheses; they are not an independent rebuild performed for this preprint.